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Final project:
Yacht Hydrodynamic Resistance

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1 Data and Problem

We analyse the *Yacht Hydrodynamics* dataset from the UCI Machine Learning Repository [1, 2]: 22 sailing-yacht hull forms tested at several speeds, for $n = 308$ observations. Each row holds five hull *form factors* — `LCB`, `prismatic`, `L_disp`, `beam_draught` and `L_beam` (constant within a hull) — and the dimensionless **Froude number** $Fn = v/\sqrt{gL} \in [0.125, 0.45]$; the response is the residuary resistance per unit weight of displacement (Table 3).

For slender hulls, physics predicts a power law resistance $\propto Fn^{\gamma_F}$ with $\gamma_F \approx 4$. We study it in a Bayesian framework:

1. log–log regression to estimate and *verify* $\hat{\beta}_F \approx 4$, with a prior-sensitivity check;
2. BAS/BMA over the 2^5 hull-factor submodels, to see which carry signal once Fn is included;
3. a heteroscedasticity-corrected nonlinear model on the retained factors, contrasted with the log–log fit;
4. posterior predictive curves, with and without the heteroscedasticity treatment, against the data.

2 Log–log Regression

2.1 Model and prior

Taking logarithms turns the physical power law resistance $\propto Fn^{\gamma_F}$ into a linear model in which the exponent appears directly as a regression slope. We therefore fit

$$\log(\text{resistance}) = \beta_0 + \beta_F \log(\text{Froude}) + \sum_{k=1}^5 \beta_k \text{hull}_k + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where hull_k denotes the five geometric form factors; the smallest observed resistance is $0.01 > 0$, so the logarithm is well defined for every record and no offset is required. As a baseline we adopt the non-informative **Jeffreys reference prior**

$$\pi(\boldsymbol{\beta} \mid \sigma^2) \propto 1, \quad \pi(\sigma^2) \propto \sigma^{-2}, \quad (2)$$

which lets the data dominate the inference; the sensitivity of the conclusions to an informative prior is examined in section 2.4.

2.2 Posterior and credible intervals

With the Jeffreys reference prior in Eq. (2), the Gaussian linear model has a closed-form posterior. Let $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$. Then

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y} \sim \mathcal{N}\left(\hat{\boldsymbol{\beta}}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\right), \quad (3)$$

Integrating out σ^2 gives a multivariate Student- t posterior for $\boldsymbol{\beta}$ with $n - p$ degrees of freedom, so the 95% credible intervals coincide numerically with the classical OLS confidence intervals.

2.3 Results

Table 4 (in the appendix) reports the posterior summaries. The model explains the data very well ($R^2 = 0.97$, residual standard deviation $s = 0.32$ on the log scale) and the posterior of the Froude exponent is sharply concentrated,

$$\beta_F \approx 4.69, \quad 95\% \text{ CI} = [4.60, 4.78],$$

confirming the expected power-law behaviour resistance $\sim Fn^4$ and verifying the project requirement $\hat{\beta}_F \approx 4$.

Fig. 1 (appendix) shows the log-log scatter with the posterior-mean line and its very tight 95% credible band (left), and the posterior of β_F (right): the whole posterior mass sits above the nominal value 4 ($\Pr(\beta_F > 4 \mid \mathbf{y}) \approx 1$), so the empirical exponent is best described as *slightly steeper* than the textbook value of four.

2.4 Sensitivity check: independent informative priors

The reference-prior analysis *verifies* $\beta_F \approx 4$ a posteriori; a complementary check *encodes* the physics in the prior and asks how strongly the prior must insist on 4 before it overrides the data. We replace Eq. (2) with **independent proper priors**,

$$\beta_F \sim \mathcal{N}(4, \tau^2), \quad \beta_j \sim \mathcal{N}(0, 100^2) \quad (j \neq F), \quad \sigma^2 \sim \text{Inv-Gamma}(10^{-3}, 10^{-3}), \quad (4)$$

all components mutually independent. Since $\boldsymbol{\beta}$ and σ^2 are now independent a priori, unlike in the conjugate Normal–Inverse-Gamma family, the joint posterior is no longer available in closed form, but both full conditionals remain standard,

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y} \sim \mathcal{N}(\mathbf{V}(\mathbf{S}_0^{-1}\mathbf{m}_0 + \mathbf{X}^\top \mathbf{y}/\sigma^2), \mathbf{V}), \quad \mathbf{V} = (\mathbf{S}_0^{-1} + \mathbf{X}^\top \mathbf{X}/\sigma^2)^{-1}, \quad (5)$$

$$\sigma^2 \mid \boldsymbol{\beta}, \mathbf{y} \sim \text{Inv-Gamma}(a_0 + \frac{n}{2}, b_0 + \frac{1}{2}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2), \quad (6)$$

where $\mathbf{m}_0$ and $\mathbf{S}_0$ collect the prior means and variances of Eq. (4). We therefore sample the posterior with a **Gibbs sampler** (10 000 iterations, 2 000 burn-in; the effective sample size for β_F exceeds 4 000 in every run) and sweep the prior strength $\tau \in \{0.05, 0.1, 0.2, 0.5\}$.

Table 5 (appendix) summarises the sweep; the posterior densities are overlaid in Fig. 4 (appendix). A moderately informative prior ($\tau = 0.5$) reproduces the reference result. Tightening the prior pulls the posterior mean towards 4, but only the near-dogmatic $\tau = 0.05$ (prior 95% interval $\approx [3.9, 4.1]$) drags it as far as 4.33, and even then $\Pr(\beta_F > 4 \mid \mathbf{y}) = 1$.

For any moderately diffuse physical prior ($\tau \geq 0.1$) the likelihood dominates and the posterior mean stays in 4.56–4.68. The conclusion that the empirical exponent is genuinely steeper than Fn^4 is therefore **robust to the prior**: it is a feature of the $n = 308$ observations, not an artefact of the non-informative choice, a transparent instance of prior–data conflict, in which an informative likelihood overwhelms all but an unreasonably confident prior.

3 BAS/BMA model selection over hull covariates

The first part establishes the dominant physical relationship: after taking logs, the resistance is essentially a power law of the Froude number. The second question is conditional on that fact: once $\log(Fn)$ is already in the model, which hull-geometry variables still carry independent signal?

We address this with Bayesian model averaging using `bas.lm`. The model space contains the five optional geometry covariates `LCB`, `prismatic`, `L_disp`, `beam_draught` and `L_beam`. Therefore the number of candidate submodels is only $2^5 = 32$. The term $\log(Fn)$ is forced into every model, because the project is not asking whether speed matters; it is asking which hull covariates matter *after* speed has been accounted for. In the R code this is enforced with `include.always = ~ log_Fr`, while the five hull terms are left optional.

The main diagnostic is the posterior inclusion probability (PIP). A PIP is the posterior probability that a covariate is included in the true regression model, after averaging over all 32 submodels. Hence a large PIP indicates that the variable repeatedly appears in high-posterior-probability models, while a small PIP means that the data do not strongly need that variable once the other terms are considered.

Using the BIC approximation to the marginal likelihood, the highest posterior model is the one containing $\log(Fn)$ and `beam_draught`, but its posterior probability is not dominant. The approximate PIPs are reported in Table 1. The largest value is for `beam_draught`, but even there the evidence is only moderate. The remaining covariates have weak inclusion probabilities. Thus the BAS/BMA conclusion is deliberately cautious: the Froude power law is the stable signal; the hull-form variables may have secondary effects, but no individual hull covariate is strongly identified by the homoscedastic log-log regression.

This agrees with the full-model coefficient intervals in Table 4, where all five hull coefficients have 95% credible intervals crossing zero. BAS/BMA strengthens that interpretation because it checks the whole model space instead of relying on one full regression model.

Table 1: Approximate posterior inclusion probabilities for the five optional hull-geometry covariates when $\log(Fn)$ is forced into every model. Values are based on enumeration of the $2^5 = 32$ submodels using a BIC marginal-likelihood approximation, matching the `prior = "BIC"` specification in `bas.lm`.

Covariate	PIP	Interpretation
<code>beam_draught</code>	0.57	moderate, not decisive
<code>prismatic</code>	0.27	weak
<code>LCB</code>	0.21	weak
<code>L_disp</code>	0.21	weak
<code>L_beam</code>	0.19	weak

4 Heteroscedasticity-Corrected Nonlinear Model

5 Model Comparison

We compare the candidate models using two complementary criteria. First, WAIC and PSIS-LOO compare expected out-of-sample predictive accuracy from the pointwise log-likelihood; lower WAIC/LOOIC is better. Second, posterior predictive coverage checks whether the predictive bands actually contain the observed data with reasonable width.

This is a Bayesian model comparison because all quantities are computed from the posterior distribution, not from a single point estimate. For each candidate model we use MCMC draws of the parameters to evaluate the pointwise posterior log-likelihood $\log p(y_i | \theta^{(s)})$. WAIC estimates predictive accuracy by combining the log posterior predictive density with a penalty for the effective number of parameters, while PSIS-LOO approximates leave-one-out cross-validation using the same posterior draws. The posterior predictive coverage is also Bayesian: the bands are obtained by simulating replicated data y^{rep} from the posterior predictive distribution $p(y^{\text{rep}} | y)$ and then checking how well those predictive intervals agree with the observed yacht-resistance data.

The first comparison asks whether the hull covariate selected by BAS/BMA, `beam_draught`, improves the nonlinear model once Froude is already in the mean. The Froude-only nonlinear model (M0) gives WAIC = 847.8 and LOOIC = 847.8, whereas the model with Froude and `beam_draught` (M1) gives WAIC = 842.4 and LOOIC = 842.5. Thus M1 is better by about 5.4 WAIC units and 5.3 LOOIC units. The PSIS-LOO comparison gives $\text{elpd}_{\text{diff}} = -2.7$ for M0 relative to M1, with standard error 3.2, so the improvement is moderate rather than overwhelming. Still, the direction agrees with the posterior coefficient result: adding `beam_draught` improves the model and the coefficient is negative.

The second comparison is more decisive. Keeping the same M1 mean structure, we compare the constant-variance additive model with the learned-variance model, where the residual scale is allowed to depend on the fitted mean. The constant-variance model gives WAIC = 1166.8 and LOOIC = 1167.2, while the learned-variance model gives WAIC = 842.4 and LOOIC = 842.5. The differences, $\Delta\text{WAIC} = 324.3$ and $\Delta\text{LOOIC} = 324.7$, are very large. PSIS-LOO gives an elpd difference of -162.3 for the constant-variance model relative to the learned-variance model, with standard error 21.7. This is strong evidence that modelling the heteroscedasticity is necessary.

Table 2: Information-criterion comparison of the nonlinear models. Lower WAIC and LOOIC are better.

Model	WAIC	LOOIC	ΔWAIC	ΔLOOIC
M0: Froude only	847.8	847.8	5.4	5.3
M1: Froude + <code>beam_draught</code>	842.4	842.5	0.0	0.0
M1: constant variance	1166.8	1167.2	324.3	324.7
M1: learned variance	842.4	842.5	0.0	0.0

The posterior predictive results tell the same story. The constant-variance M1

model covers about 94.5% of the observations with an average band width of 6.179. The learned-variance model covers about 98.4% of the observations with a narrower average band width of 5.535. Therefore, the learned-variance model is not merely wider; it allocates uncertainty more efficiently, narrowing the band where low-Froude observations are tightly concentrated and widening it where high-Froude observations are more dispersed.

Overall, the best model is the nonlinear M1 model with `beam_draught` and learned heteroscedastic variance. It gives the strongest predictive performance by WAIC/PSIS-LOO, better calibrated posterior predictive bands, and a physically interpretable conclusion: Froude remains the dominant driver, while `beam_draught` contributes secondary hull-geometry information once the variance structure is modelled correctly.

References

- [1] I. Ortigosa, R. Lopez, and J. Garcia. *Yacht Hydrodynamics*. UCI Machine Learning Repository. DOI: 10.24432/C5XG7R. 2007.
- [2] J. Gerritsma, R. Onnink, and A. Versluis. *Geometry, Resistance and Stability of the Delft Systematic Yacht Hull Series*. Technical Report. Delft University of Technology, Ship Hydromechanics Laboratory, 1981.

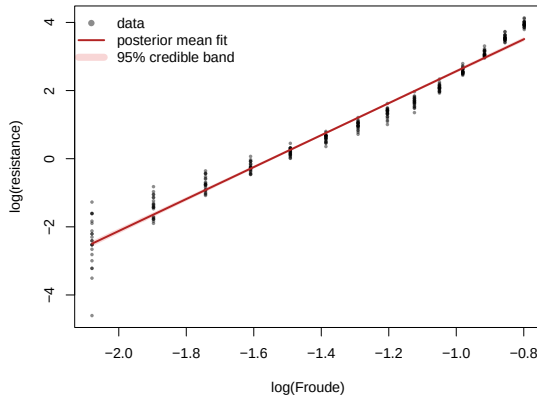
A Tables and Figures

Table 3: Variables of the yacht hydrodynamics dataset.

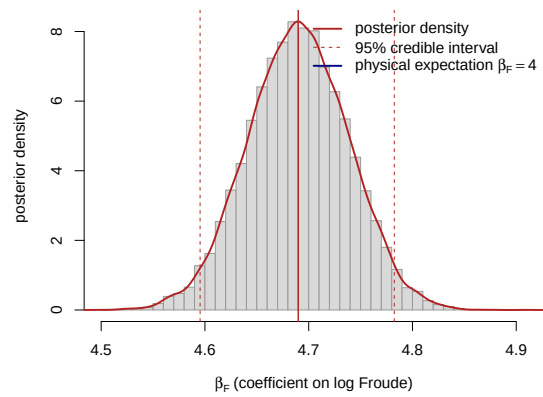
Variable	Description
LCB	Longitudinal position of the centre of buoyancy
prismatic	Prismatic coefficient
L_disp	Length–displacement ratio
beam_draught	Beam–draught ratio
L_beam	Length–beam ratio
Froude	Froude number, $Fn = v/\sqrt{gL}$
resistance	Residuary resistance per unit weight of displacement

Table 4: Posterior summary of the log–log model in Eq. (1) under the reference prior. Point estimates and 95% credible intervals coincide with the OLS fit (`lm/confint`); $\text{Pr}(> 0)$ is the posterior probability that the coefficient is positive.

Coefficient	Mean	SD	2.5%	97.5%	$\text{Pr}(> 0)$
Intercept β_0	7.183	0.983	5.248	9.119	1.00
$\log(\text{Froude})$ β_F	4.690	0.048	4.596	4.785	1.00
LCB	0.022	0.012	−0.003	0.046	0.96
prismatic	−0.621	1.601	−3.771	2.529	0.35
L_disp	0.465	0.513	−0.546	1.475	0.82
beam_draught	−0.066	0.200	−0.460	0.327	0.37
L_beam	−0.464	0.515	−1.477	0.549	0.18



(a) Power-law fit on log–log axes.



(b) Posterior of β_F .

Figure 1: Left: observed $\log(\text{resistance})$ against $\log(\text{Froude})$ with the posterior-mean regression line (hull covariates held at their means) and its 95% credible band. Right: Student- t posterior density of the Froude exponent β_F ; the dashed lines mark the 95% credible interval and the solid blue line the physical reference value $\beta_F = 4$.

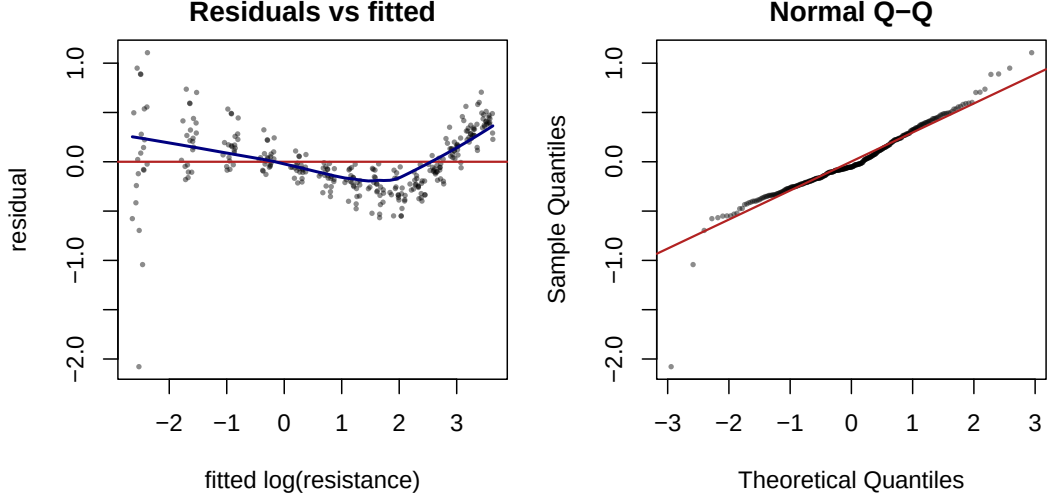


Figure 2: Residual diagnostics for the log-log model. Left: residuals versus fitted values, with a LOWESS trend (blue) revealing speed-dependent curvature and a variance that shrinks as the fitted value grows. Right: Normal Q-Q plot, showing heavier-than-Gaussian lower tails.

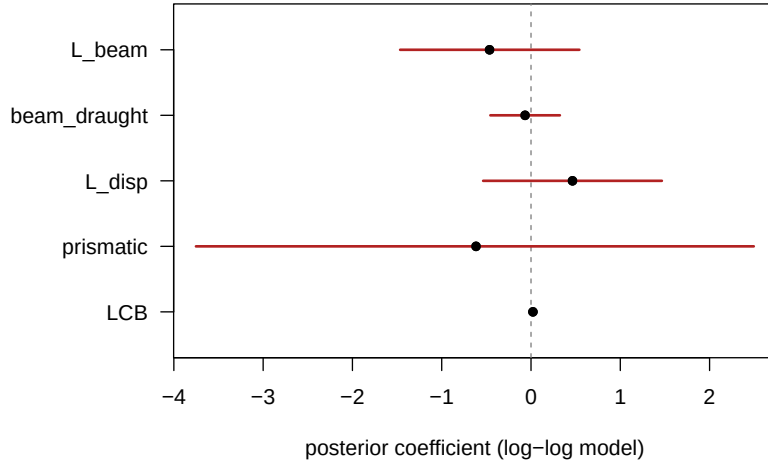


Figure 3: Posterior means and 95% credible intervals of the five hull-geometry coefficients in the log-log model. Every interval straddles zero (dashed line), indicating that no single form factor is resolved once the Froude number is accounted for.

Hull-geometry covariates. Once the Froude number is in the model, none of the five hull form factors is individually credible: every 95% interval comfortably contains zero (Table 4 and Fig. 3). The form factors vary only across the 22 hulls and are strongly collinear — the **prismatic** coefficient, for instance, has a posterior standard deviation of about 1.6. This near-redundancy is precisely what the BAS/BMA model selection in section 3 resolves, through posterior inclusion probabilities rather than single-model p -values.

Residual diagnostics. Although the global fit is excellent, the residuals (Fig. 2) reveal two systematic defects. First, a U-shaped trend against the fitted values: a

single constant exponent slightly over-predicts at intermediate speeds and under-predicts at the highest speeds. Second, the residual spread is markedly larger at low fitted values, i.e. at low Froude numbers where the resistance is tiny and dominated by measurement noise — a textbook signature of heteroscedasticity, also visible as the fanning of the data cloud on the left of Fig. 1. Both findings directly motivate the heteroscedasticity-corrected nonlinear model developed in the main text.

Table 5: Posterior of the Froude exponent β_F under the independent informative priors in Eq. (4) of increasing strength, obtained by Gibbs sampling, compared with the closed-form reference-prior result.

Prior on β_F	Mean	2.5%	97.5%	$\Pr(\beta_F > 4)$
$\mathcal{N}(4, 0.05^2)$	4.33	4.25	4.41	1.00
$\mathcal{N}(4, 0.10^2)$	4.56	4.47	4.64	1.00
$\mathcal{N}(4, 0.20^2)$	4.65	4.56	4.75	1.00
$\mathcal{N}(4, 0.50^2)$	4.68	4.59	4.78	1.00
reference $\pi(\beta, \sigma^2) \propto \sigma^{-2}$	4.69	4.60	4.78	1.00

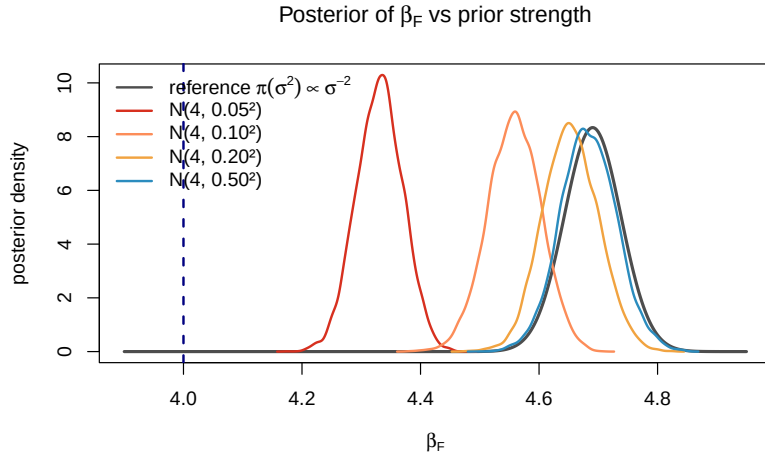


Figure 4: Posterior density of the Froude exponent β_F under the independent informative priors $\mathcal{N}(4, \tau^2)$ of increasing strength (coloured, Gibbs samples) and under the reference prior (grey, closed-form Student- t). The dashed blue line marks the physical value $\beta_F = 4$. Tightening the prior toward 4 shifts the posterior only modestly; the data keep it well above 4 unless the prior is unreasonably confident.